

WEEK 1 ASSIGNMENT

Task Name

Mathematical Customer Lifetime Value (CLV) Projection via BG/NBD Model from Scratch

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Introduction

Customer Lifetime Value (CLV) is a business metric used to estimate the total value a customer will bring to a company during their relationship with the business. It helps organizations identify valuable customers, improve marketing strategies, and increase customer retention.

The BG/NBD (Beta-Geometric/Negative Binomial Distribution) model is a statistical model that predicts future customer purchases based on past transaction history. It estimates how frequently a customer will make purchases and the probability that the customer has become inactive.

Objective

The objective of this assignment is to understand and implement the BG/NBD model from scratch without using any pre-built marketing libraries. The model estimates customer purchasing behaviour by calculating four parameters: r , α (alpha), a , and b using numerical optimization.

Tools Required

- Python
- NumPy
- SciPy

- Pandas
- Jupyter Notebook or Google Colab

BG/NBD Model Parameters

r : Represents customer purchase frequency.

α (Alpha): Controls the transaction rate of customers.

a : Represents one parameter of the customer dropout probability.

b : Represents another parameter of the customer dropout probability.

These four parameters help estimate future customer purchases and Customer Lifetime Value.

Methodology

1. Collect customer transaction data.
2. Calculate Frequency, Recency, and Customer Age (T).
3. Write the BG/NBD likelihood function mathematically.
4. Implement the likelihood function in Python.
5. Use NumPy for vectorized calculations.
6. Optimize the parameters (r , α , a , b) using SciPy's L-BFGS-B optimization algorithm.
7. Predict future customer purchases and estimate Customer Lifetime Value.

Advantages of BG/NBD Model

- Predicts future customer purchases.

- Helps identify loyal customers.
- Supports personalized marketing campaigns.
- Improves customer retention.
- Assists businesses in maximizing profit.

Applications

- E-commerce websites
- Retail stores
- Banking
- Subscription services
- Online shopping platforms

Conclusion

The BG/NBD model is an effective statistical approach for predicting customer purchasing behavior and estimating Customer Lifetime Value. By implementing the model from scratch, students gain a deeper understanding of customer analytics, probability, optimization techniques, and machine learning concepts. This knowledge helps businesses make informed decisions and improve customer relationship management.